

TORUNGEN FYR, Norway

WMO: 014650

Lat: 58.399N

Lon: 8.789E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.8	-6.9	-11.9	1.3	-6.6	-10.1	1.6	-5.0	18.2	0.9	16.5	0.5	6.8	20	0.791

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.4	22.1	18.0	21.2	17.5	20.1	16.5	19.2	21.3	18.2	20.4	17.4	19.3	5.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	20.7	17.3	12.4	19.6	16.5	11.7	18.8	54.6	21.4	51.6	20.4	49.0	19.4	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.8	13.3	-9.9	24.2	2.8	1.8	-11.9	25.5	-13.6	26.6	-15.2	27.6	-17.3	29.0		
(5)			-10.3	19.5	2.8	1.2	-12.3	20.4	-13.9	21.1	-15.5	21.7	-17.6	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	1.3	0.7	2.5	6.0	10.5	14.5	16.9	16.6	13.8	9.4	5.6	2.6
	DBStd	6.49	3.66	3.70	3.21	2.40	2.74	2.21	1.99	1.79	2.02	2.84	3.31	4.32
	HDD10.0	1319	269	259	233	123	25	1	0	0	1	46	133	229
	HDD18.3	3639	527	493	492	371	242	118	54	60	137	278	381	488
	CDD10.0	737	0	0	0	2	41	134	214	204	114	26	2	0
	CDD18.3	18	0	0	0	0	0	2	11	6	0	0	0	0
	CDH23.3	9	0	0	0	0	0	3	5	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.7	7.7	6.9	6.8	6.1	6.2	6.0	5.8	6.0	6.7	7.3	7.6	7.5
Precipitation	PrecAvg	1165	117	76	58	65	75	77	81	120	116	132	143	105
	PrecMax	1609	270	216	132	145	143	151	193	215	286	269	430	192
	PrecMin	795	22	18	8	15	12	7	15	28	36	32	45	30
	PrecStd	203	60	45	34	35	35	38	41	53	61	63	82	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	9.1	12.5	14.5	20.6	22.5	24.0	22.8	19.7	15.7	13.0	10.9
		MCWB	7.9	6.1	7.8	9.3	17.2	16.2	19.0	18.3	14.8	13.1	11.6	9.6
	2%	DB	7.8	7.4	9.2	12.1	17.7	20.8	22.4	21.8	18.4	14.7	11.9	9.8
		MCWB	6.6	5.3	5.8	8.1	13.7	15.7	18.5	18.2	14.9	12.9	10.6	8.7
	5%	DB	7.0	6.3	7.4	10.6	16.2	19.2	21.5	20.7	17.6	14.0	10.9	9.0
		MCWB	6.0	4.9	4.7	7.7	12.5	14.8	18.0	17.3	15.0	12.3	9.6	8.0
	10%	DB	6.2	5.5	6.2	9.5	14.6	17.9	20.5	19.7	17.0	13.4	10.1	8.2
		MCWB	5.2	4.2	4.5	7.1	11.6	14.4	17.2	16.5	14.8	11.8	8.9	7.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	6.8	8.3	10.5	17.7	18.5	20.6	19.7	17.2	14.6	12.0	10.0
		MCDB	8.9	7.7	11.9	12.6	19.8	20.6	23.0	21.8	17.8	15.0	12.5	10.4
	2%	WB	7.1	6.0	6.3	9.3	14.6	16.7	19.6	18.9	16.5	13.5	10.9	9.1
		MCDB	7.6	6.6	7.9	11.0	17.1	19.1	21.7	21.1	17.3	14.3	11.7	9.7
	5%	WB	6.2	5.2	5.5	8.4	13.3	15.8	18.6	18.1	15.9	12.8	10.0	8.3
		MCDB	6.8	6.2	6.8	10.1	15.4	18.1	20.8	20.2	17.0	13.6	10.8	9.0
	10%	WB	5.3	4.3	4.9	7.4	12.1	15.1	17.7	17.3	15.2	12.1	9.1	7.3
		MCDB	6.1	5.2	6.0	8.9	14.1	17.3	19.8	19.2	16.6	13.0	9.9	8.1
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.6	4.2	4.2	4.4	4.6	4.4	4.4	4.3	3.9	3.7	3.8
		MCDBR	4.1	4.4	6.1	6.2	6.1	6.6	5.7	5.4	4.8	4.1	4.2	3.8
		MCWBR	4.0	3.5	3.9	4.1	4.0	4.2	3.7	3.6	3.7	3.6	4.4	3.9
	5% WB	MCDBR	3.8	3.9	4.9	5.0	5.1	4.8	4.6	4.3	3.9	3.5	4.0	3.6
		MCWBR	4.0	3.4	3.4	3.6	3.7	4.0	3.5	3.4	3.5	3.6	4.3	3.9
Clear-Sky Solar Irradiance	taub	0.295	0.325	0.343	0.382	0.373	0.358	0.378	0.379	0.354	0.338	0.308	0.273	
	taud	2.239	2.317	2.362	2.317	2.415	2.481	2.441	2.436	2.495	2.440	2.297	2.184	
	Ebn at Noon	534	701	801	824	861	880	850	822	788	684	514	430	
	Edh at Noon	47	70	88	108	105	99	102	95	77	62	44	35	
All-Sky Solar Radiation	RadAvg	0.34	0.89	2.32	3.90	5.28	6.00	5.49	4.22	2.72	1.24	0.45	0.23	
	RadStd	0.08	0.19	0.21	0.54	0.58	0.43	0.53	0.26	0.32	0.19	0.11	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-144	N/A	N/A

Nomenclature: See separate page